



Qawqa'a: Auditory Habilitation System for Children with Cochlear Implants Using Virtual Reality

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ABSTRACT

Qawqa'a is a VR-based game for accessible and engaging auditory habilitation in Arabic, targeted for children from 8 to 12 years old with bilateral (both ears) cochlear implants. It serves as a digital home-training kit consisting of two parts, a game side for the child for improving auditory skills for sound localization and speech perception, and a web-based monitoring and tracking side for habilitation center administrators, facilitating efficient tracking of a child's progress and skills within the game.

Qawqa'a holds significant promise in addressing the challenges faced by families in accessing regular habilitation sessions. By providing a convenient, cost-effective, and engaging habilitation experience, aiming to improve adherence to treatment protocols, thereby maximizing the benefits of cochlear implants.

PROBLEM

Habilitation after cochlear implant surgery is essential for individuals with severe to profound hearing loss. However, accessing auditory habilitation and speech therapy poses challenges, including availability, cost, and accessibility. These obstacles restrict their ability to maximize the benefits of cochlear implants. To address these challenges, we have developed our project, which offers innovative solutions such as digital home-training. These enable remote access to essential therapy, ensuring that children can receive the necessary support to develop their auditory and communication skills effectively.

SOLUTION

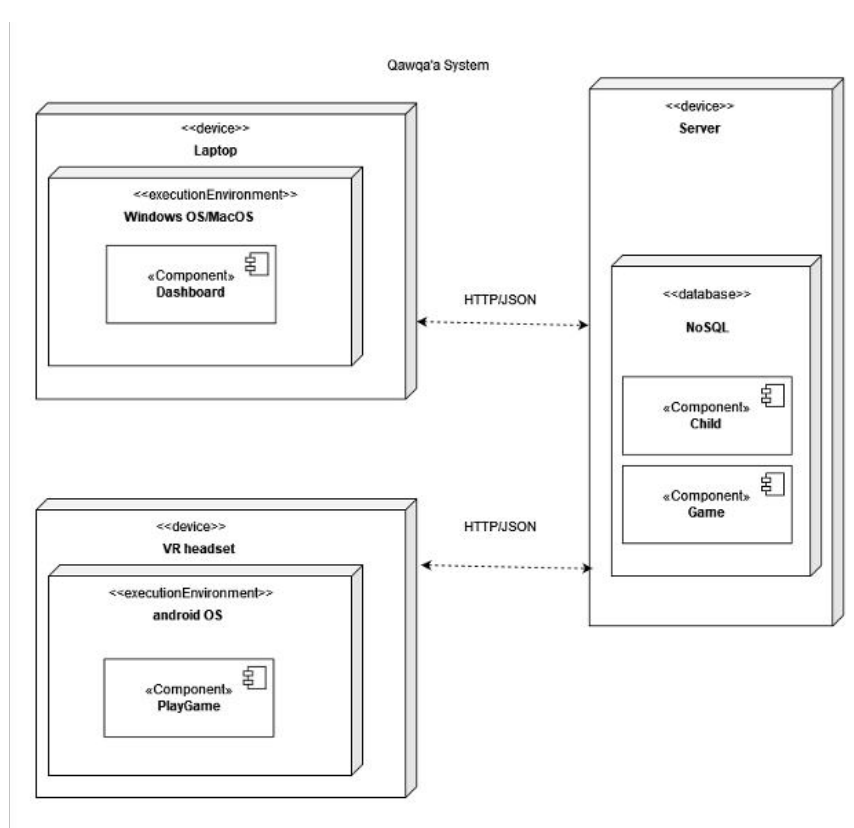
Our solution involves the development of a VR-based hearing training game in Arabic tailored for young individuals with bilateral cochlear implants. This innovative game aims to overcome accessibility challenges to habilitation services by offering engaging and immersive training experiences. It focuses on enhancing hearing and language skills, including noise perception, sound localization, speech understanding, and verbal intelligence. By targeting various abilities, reducing clinic visits, and transforming training into an enjoyable experience.

OBJECTIVES

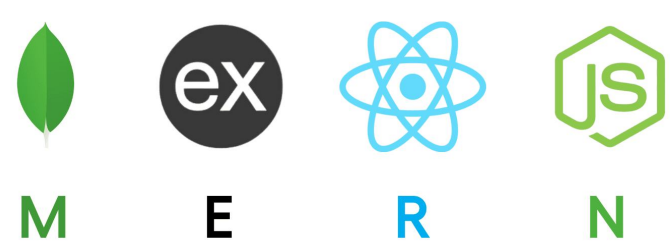
- Facilitate an accessible auditory habilitation
- Empower children's skill development consistently
- Enhance child engagement and motivation
- Enabling effective progress monitoring

ARCHITECTURE

In Qawqa'a system, three interconnected nodes drive remote habilitation services for children with cochlear implants: the Database Server, Admin's Computer, and Child's VR Headset. The Database Server stores child and game data, the Admin's Computer manages user registration and progress, and the Child's VR Headset delivers immersive experiences via the PlayGame module.

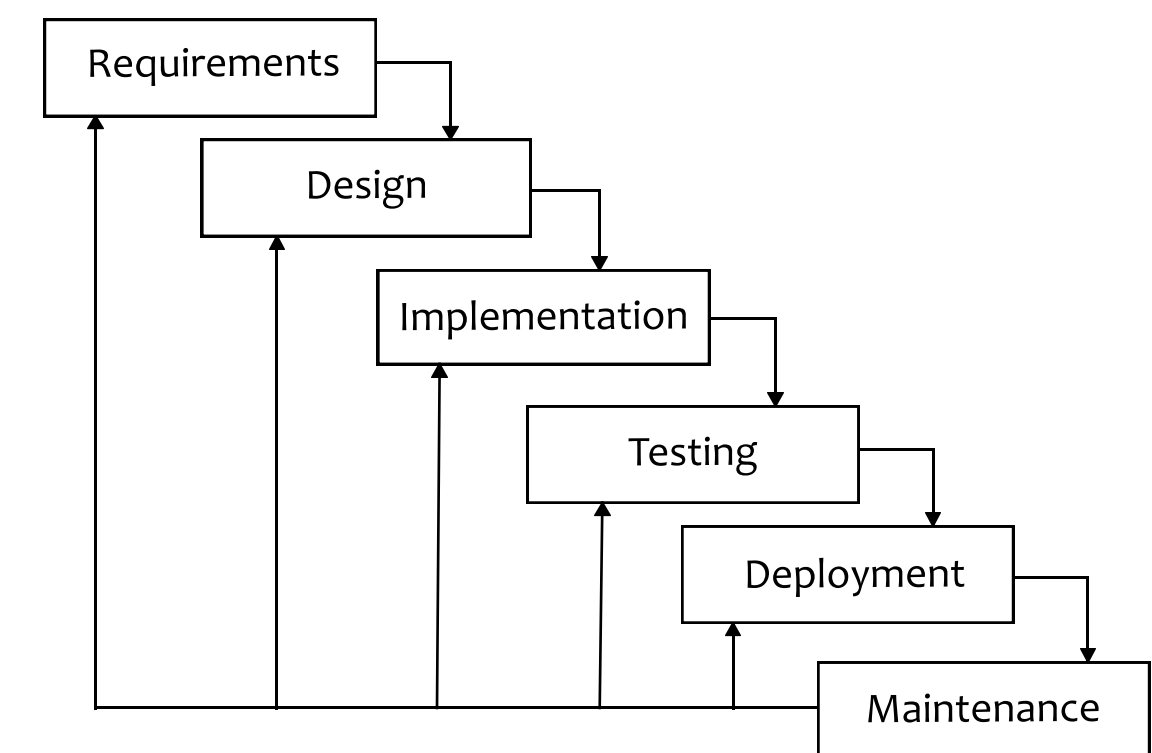


TOOLS



METHODOLOGY

Waterfall Methodology

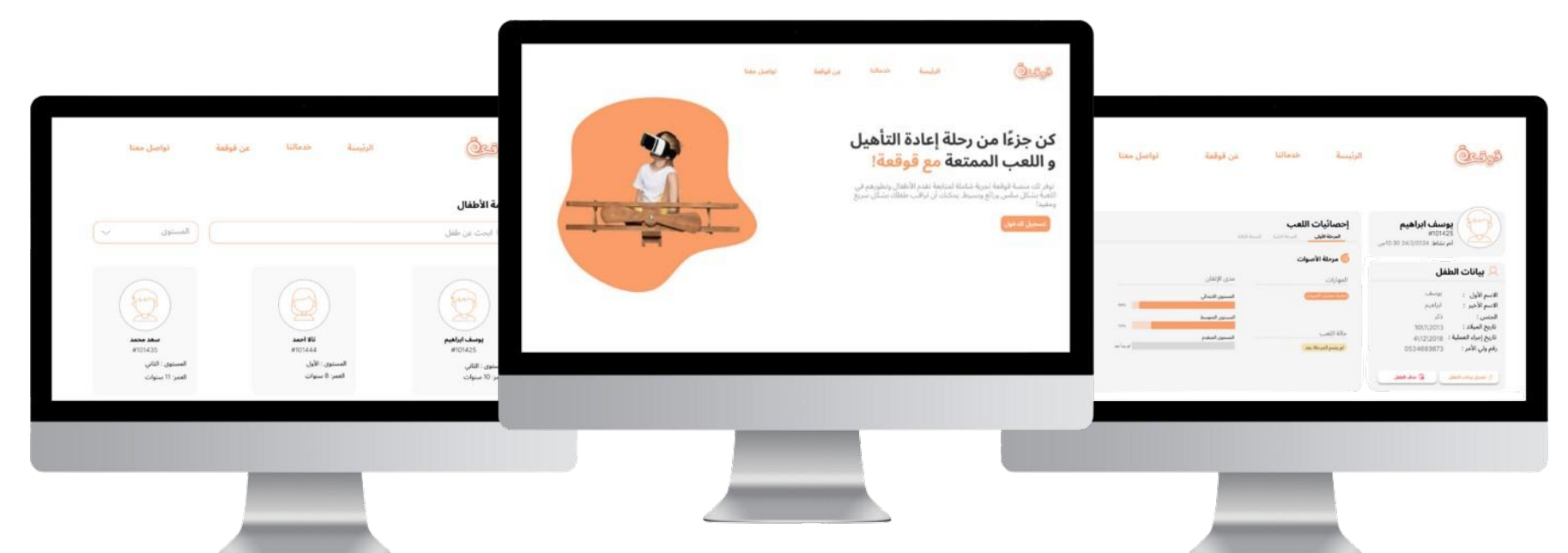


RESULTS

VR Side



Web Side



CONCLUSION

Qawqa'a has been designed to support the cochlear implant community by facilitating remote access to habilitation services for every child. , regardless of any limitations. This means that every child with a cochlear implant can keep up with their training, overcoming any obstacles that might have held back their progress before. It also empowers habilitation center administrators to monitor each child's progress, while turning the training process into an engaging and enjoyable experience.



Know more about Qawqa'a